



PHILOSOPHY OF SCIENCE

Lecture 4: Scientific Explanation and Scientific Understanding

Enrico Cinti

Tuesday, 22 September 2026



OUTLINE

(A) Scientific explanation

- Hempel's deductive-nomological model
- Critiques and alternative models

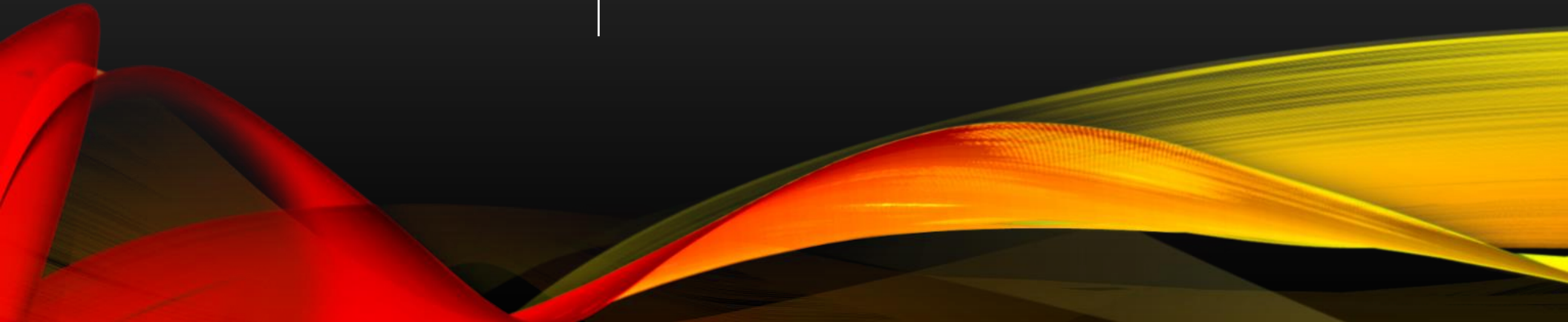
(B) Scientific understanding

- Background views about understanding
- Pragmatic theories of understanding
- De Regt & Dieks' pragmatic theory of understanding

Hempel's deductive-
nomological model

Critiques and alternative models

(A) SCIENTIFIC EXPLANATION



SCIENTIFIC EXPLANATION IN PHILOSOPHY OF SCIENCE

- **Before 1948:** idea that science provides **explanations of phenomena** not a philosophical topic for logical empiricists.
 - **Aristotle's** theory of **causes** is about the structure of explanations (Moravcsik)
 - **Duhem:** explanation associated with metaphysics, “gives hostages to fortune”. Science dependent on metaphysics.
 - ‘Atoms exist’ and ‘there is an aether’ are metaphysical claims that science can neither confirm nor refute
 - **Carnap:** No ‘metaphysical causes/why-questions’ like Hegel (pseudo-explanations). Admits “empiricist explanation”:
 - **Logical empiricist** explanation closely linked with **prediction from laws**.
- **1948:** Carl Hempel and Paul Oppenheim: ‘covering law model’
 - “Philosophically light” (epistemic) account of explanation

HEMPEL'S DEDUCTIVE-⁵ NOMOLOGICAL MODEL

- **Explanation** = deductive argument in which the *explanandum* (phenomenon or law to be explained) is deduced from premises containing general laws and particular facts and initial/boundary conditions (*explanans*):

Explanans:

$C_1, C_2, \dots, C_k$ particular facts/conditions
 $L_1, L_2, \dots, L_r$ general laws

Explanandum:

E phenomenon

COVERING LAW MODEL

- **Deductive-nomological (D-N) variant:**
- **Explanation** = deduction from empirical content and general laws
 - **Deductive:** from premises to logically reach a conclusion
 - **Nomological:** use of 'laws' of nature (Greek 'nomos')
- **Conditions** of adequacy:
 1. Explanandum must be **logical consequence** of explanans
 2. Explanans must contain a **general law** in an **essential** way (i.e. indispensable)
 3. Explanans must have **empirical content**: it must be capable, at least in principle, of test by experiment/observation
 4. Explanans must be **true**
- **Inductive-statistical (I-S):** at least one probabilistic law
 - Explanandum is a true singular fact whose **high probability** is shown.

EXAMPLE

Lately, I have difficulties opening and closing my front door. I ask myself, why?



EXAMPLE

Lately, I have difficulties opening and closing my front door. I ask myself, why?

Explanation:

My front door is made of wood

Wooden objects expand if humidity increases

Humidity has increased this autumn

My front door has expanded this autumn

Hempel: symmetry between explanation and prediction. They have the same logical structure



- | Group | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 |
|----------|----------|----------|----------|----------------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|
| Period 1 | 1
H | | | | | | | | | | | | | | | | | 2
He |
| Period 2 | 3
Li | 4
Be | | | | | | | | | | | 5
B | 6
C | 7
N | 8
O | 9
F | 10
Ne |
| Period 3 | 11
Na | 12
Mg | | | | | | | | | | | 13
Al | 14
Si | 15
P | 16
S | 17
Cl | 18
Ar |
| Period 4 | 19
K | 20
Ca | 21
Sc | 22
Ti | 23
V | 24
Cr | 25
Mn | 26
Fe | 27
Co | 28
Ni | 29
Cu | 30
Zn | 31
Ga | 32
Ge | 33
As | 34
Se | 35
Br | 36
Kr |
| Period 5 | 37
Rb | 38
Sr | 39
Y | 40
Zr | 41
Nb | 42
Mo | 43
Tc | 44
Ru | 45
Rh | 46
Pd | 47
Ag | 48
Cd | 49
In | 50
Sn | 51
Sb | 52
Te | 53
I | 54
Xe |
| Period 6 | 55
Cs | 56
Ba | 57
La | *
72
Hf | 73
Ta | 74
W | 75
Re | 76
Os | 77
Ir | 78
Pt | 79
Au | 80
Hg | 81
Tl | 82
Pb | 83
Bi | 84
Po | 85
At | 86
Rn |
| Period 7 | 87
Fr | 88
Ra | 89
Ac | *
104
Rf | 105
Db | 106
Sg | 107
Bh | 108
Hs | 109
Mt | 110
Ds | 111
Rg | 112
Cn | 113
Nh | 114
Fl | 115
Mc | 116
Lv | 117
Ts | 118
Og |
| | | | | *
58
Ce | 59
Pr | 60
Nd | 61
Pm | 62
Sm | 63
Eu | 64
Gd | 65
Tb | 66
Dy | 67
Ho | 68
Er | 69
Tm | 70
Yb | 71
Lu | |
| | | | | *
90
Th | 91
Pa | 92
U | 93
Np | 94
Pu | 95
Am | 96
Cm | 97
Bk | 98
Cf | 99
Es | 100
Fm | 101
Md | 102
No | 103
Lr | |

COMMENTS ABOUT HEMPEL'S MODEL

- Laws do not need to be causal:
 - **Functional laws admitted**
- Model is **normative**:
 - Not simply meant to **describe** how scientists actually explain, but ideal of **good scientific explanation**
 - Similar to concept of 'proof' by logicians and philosophers of mathematics
 - **Elliptically formulated explanation**: scientists often give **incomplete** explanations

OBJECTIONS TO D-N MODEL

Asymmetry of explanations, but
(sometimes) symmetry of deductions

Irrelevance: may add irrelevant
information that rob the argument
of explanatory power

Every sample of sodium chloride crystals
hexed by a magician dissolves in water
(Salmon and Achinstein)

ASYMMETRY OF EXPLANATION

- The following deduction goes both ways, i.e. we can use trigonometry to interchange 'flagpole' 'shadow' to predict, but not (always) to explain:

Laws of optics

Laws of geometry

Position of the sun

Length of the flagpole

Length of the shadow

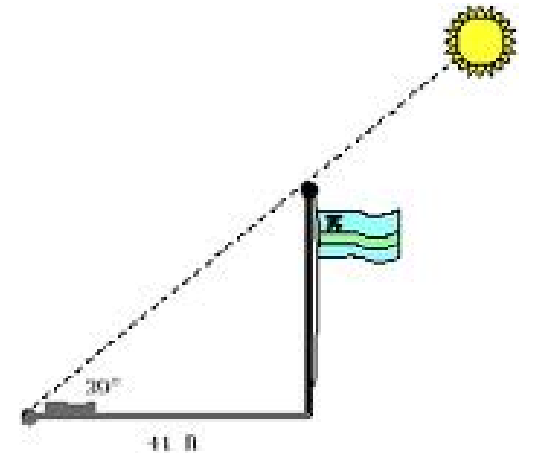
Laws of optics

Laws of geometry

Position of the sun

Length of the shadow

Length of the flagpole



BAROMETER EXAMPLE

If the barometer drops, a storm will occur

The barometer drops

A storm will occur

- **Correlation** $\neq$ explanation



ALTERNATIVE THEORIES OF EXPLANATION

1. Explanation related to **causation**: good explanations show what caused the phenomena
2. Explanation related to **unification**: good explanations show how the phenomena fit into a broader pattern
3. Explanation is **pragmatic**: it depends on the *context* whether answers to why-questions are satisfactory explanations
4. Distinguish **scientific explanation** (require scientific theory) from explanations in daily life

CAUSAL THEORIES OF EXPLANATION

- **Explanation** = uncovering the **causes** of phenomena
 - Causal processes, causal interactions, and causal laws provide the **mechanisms** by which the world works; to understand why certain things happen, we need to see how they are produced by these mechanisms (W. Salmon, 1984)
- **What is causality?**
 - **Regularity theory (Hume)**: causation is **constant conjunction**: C regularly followed by E. Produces **expectation**, not necessary connection in nature.
 - **Counterfactual theories (Hume)**: based on intuition that **if the cause C hadn't occurred, the effect E wouldn't have occurred** (J. Woodward)
 - **Process theories**: based on the idea that one can identify **causal processes** and interactions in nature (W. Salmon)



CAUSAL-MECHANICAL EXPLANATION (SALMON)

- Explanation situates explanandum in **network or nexus of causal relations i.e. the causal structure of the world**: to explain is to systematically causally relate explanandum to other items.
 - 'The water evaporated because heat was applied to it, and the van der Waals bonds between the molecules were broken'
 - **Exhibits the causal structure**
- Giving an explanation is answering a why-question with 'because...', hence citing the causes for the explanandum.
- How do we distinguish causation from correlation? (Hume's problem)
- Too narrow? Perhaps not all explanations need to invoke causation (De Regt & Dieks).

UNIFICATIONIST THEORIES OF EXPLANATION

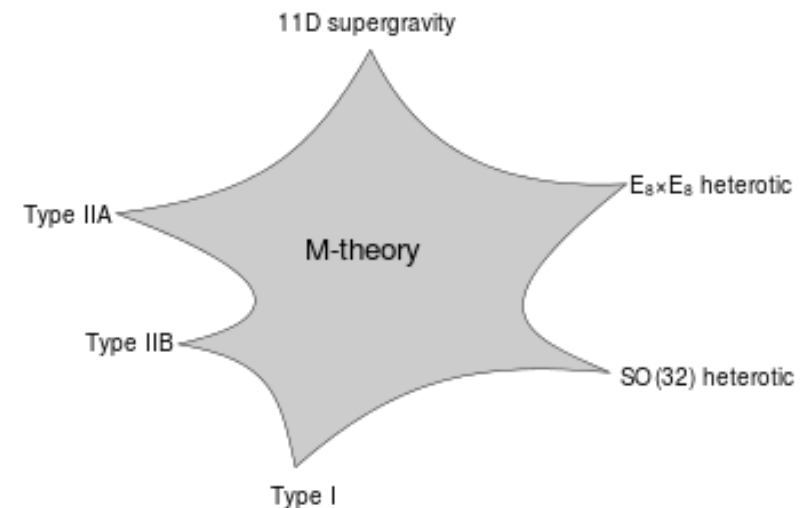
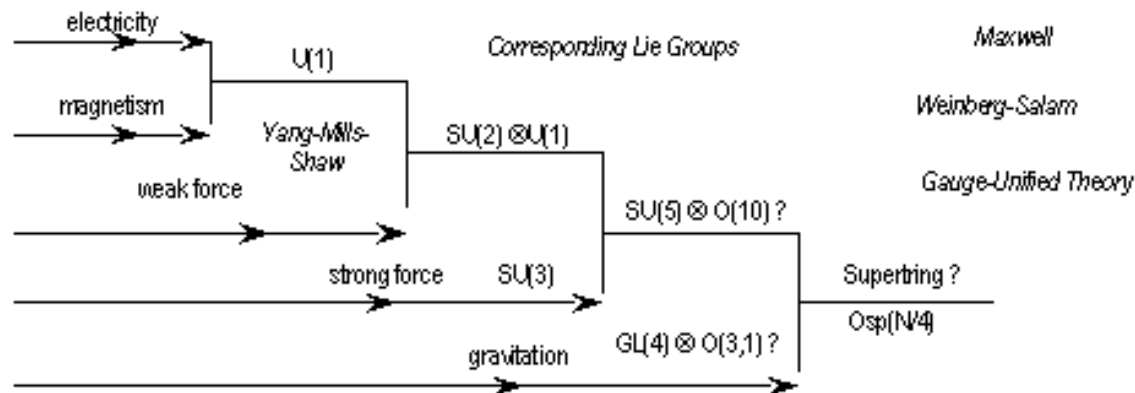
- **Unification:** phenomena are explained by fitting them into broader pattern
 - Science advances our understanding of nature by showing us how to derive descriptions of many phenomena, using the same patterns of derivation again and again, and, in demonstrating this, it teaches us how to **reduce the number of types of facts we have to accept as ultimate** (or brute) (P. Kitcher 1989).
- **Advantages:**
 - Accounts for non-causal explanations, e.g. explanations in quantum theory

UNIFICATION (KITCHER 1989)

- ‘Understanding the phenomena not simply a matter of reducing the “fundamental incomprehensibilities” but of seeing **connections, common patterns**, in what initially appeared to be different situations.’
- The idea is that an explanation is a certain type of reasoning or **argumentative pattern**: an argument that proceeds from some simple, unified principles, to a multiplicity of (possibly miscellaneous) events.

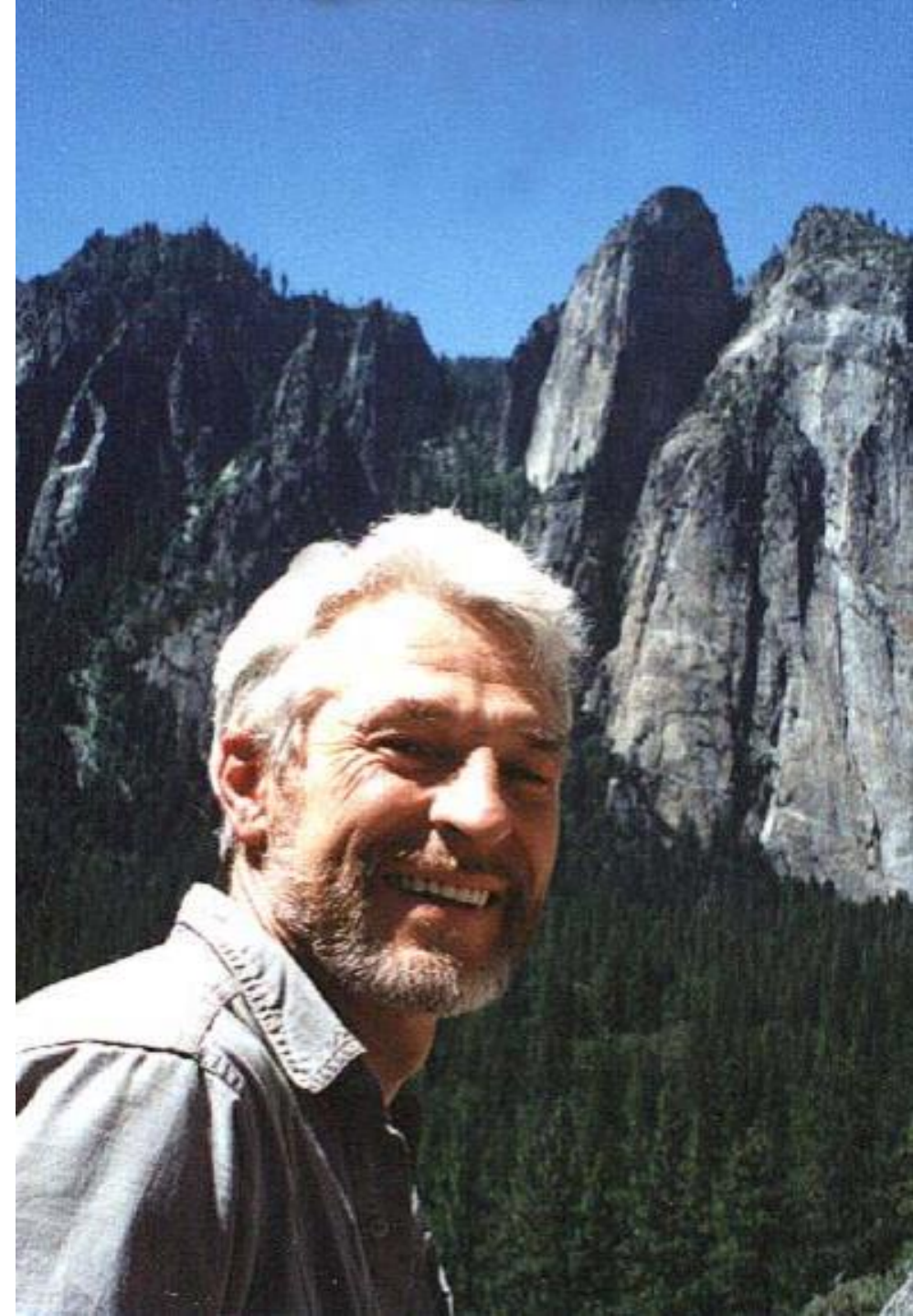
How Gauge Theory Unify the Fundamental Forces of Nature

Source: nfm



BAS VAN FRAASSEN

- Explanation/understanding: part of **pragmatic** dimension of science
 - **Contextual:** aims and preference
- Explanation is part of the **reasons we may have to accept** a theory as useful for particular purposes (e.g. make predictions, answer questions)
 - Explanations do not add to our **beliefs about the relation between the theory and the world**
 - To explain is **not an epistemic aim of science:** pragmatic dimension of **theory acceptance**



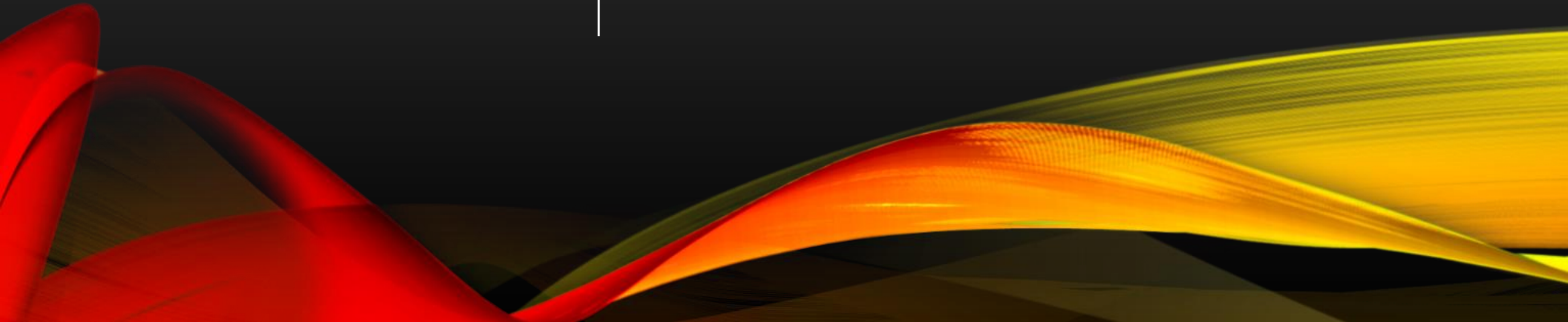
VAN FRAASSEN (1980)

- **Duhem..** argued that explanation is not an aim of science. In retrospect, **he fostered that explanation-mysticism which he attacked..** He argued that only metaphysical theories explain, and that metaphysics is.. foreign to science. Fifty years later, **Quine** having argued that there is **no demarcation between science and philosophy**, and the difficulties of the ametaphysical stance for the positivist philosophies having made a return to metaphysics tempting, one noticed that **scientific activity does involve explanation**, and **Duhem's argument was deftly reversed.**
- Once you decide that explanation is something irreducible and special, the door is opened to elaboration by means of further concepts pertaining thereto, all equally irreducible and special.. **Not everyone has joined this return to essentialism or neo-Aristotelian realism, but some eminent realists have** publicly explored or advocated it.
- The discussion of explanation went wrong at the very beginning when explanation was conceived of as a relationship like description: a relation between theory and fact. Really it is a **three-term relation, between theory, fact, and context.** An explanation is an answer.. So **scientific explanation is not (pure) science but an application of science. It is a use of science to satisfy certain of our desires.**

Background views about
understanding

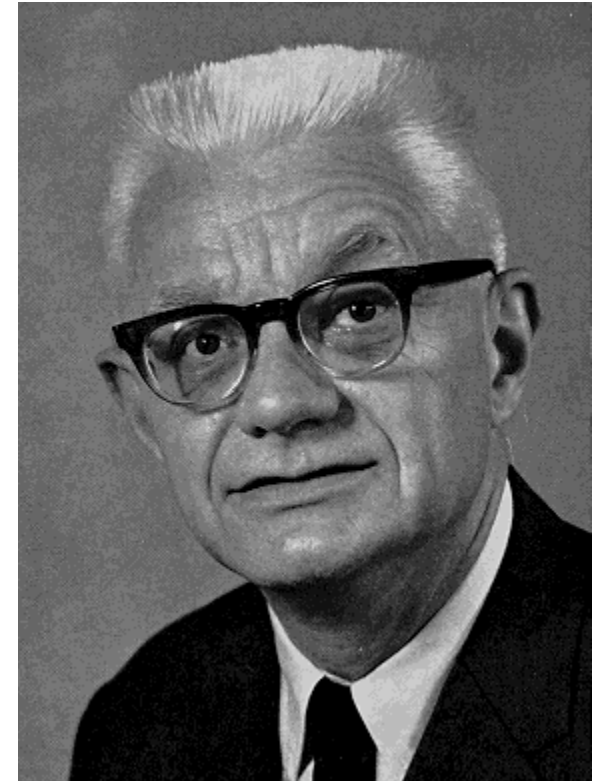
Pragmatic theories of
understanding

(B) SCIENTIFIC UNDERSTANDING



HEMPEL'S ELIMINATIVISM ABOUT UNDERSTANDING

- “Such expressions as ‘realm of understanding’ and ‘comprehensible’ do not belong to the vocabulary of logic, for they refer to the psychological and pragmatic aspects of explanation.”
 - Explanation is objective relation between **T** and **P**
 - Understanding by **S** epistemically irrelevant
- ‘Pragmatic’ used synonymously with ‘subjective’
- An investigation of understanding gives us knowledge of people’s preferences/interests not of the topic
 - Of interest to psychology not philosophy



REDUCTIVISM ABOUT UNDERSTANDING

- **Explanation is understanding enough:** no need for separate theory of understanding
- Khalifa (2012): understanding is a **form of knowledge**, reducible to explanation and cognate notions
 - Nevertheless **philosophically interesting**
- Lipton (2004): 'understanding is not some sort of super-knowledge, but simply more knowledge: knowledge of causes'
- Trout (2002): understanding as *Eureka!*-experience is the result of cognitive biases, e.g. overconfidence and hindsight

CARNAP (1939, 1947)

- 'Understanding' is a vague word that requires explication. Three senses:
 1. **Pragmatic: capability of use** for the description/prediction of facts (legitimate) (1939)
 2. **Subjective/metaphysical: intuitive understanding** (rejected) (1939)
 3. **Epistemic:** to understand a language system is to **know the semantic rules/truth conditions** (legitimate) (1947)
 - Epistemic because it says what one must **know** to count as understanding



ON THE USES OF 'PRAGMATIC' AND 'SUBJECTIVE'

- Friedman (1974), Woodward and Ross (2021): equivocation of two meanings of 'pragmatic' in Hempel/van Fraassen. Distinguish:
 1. **'Subjective'**
 2. **'Use(able) for certain aims'**: compatible with objective (inter-subjective) notion of understanding
- On the second notion, understanding can be an **epistemic aim of science**, achieved by adequate explanations
- Question: what is the relation between **explanation and understanding**?

PRAGMATIC THEORIES OF UNDERSTANDING

- Requirements for understanding common to various theories:
 - (1) Explanation:** there is a scientific explanation of the phenomenon.
Explanation often in Hempel's sense
 - Bridge law between phenomena and theory: deduction/argument/model
 - (2) Requirements of adequacy:** theoretical virtues/epistemic values, usually: internal consistency and empirical confirmation (also: approximate truth)
 - (3) Useability:** it should be possible to construct adequate explanations
- Authors who reduce understanding to explanation accept only (1) and (2):
understanding = adequate explanation = knowledge
- Authors for whom understanding is 'more than' explanation add (3).
 - Thus discussion focusses on (3)
 - (3) is not about *knowledge* but *ability to act/do*: involves skills, judgement

REQUIREMENT (3) AND OBJECTIVITY

- To emphasize the objectivity of requirement (3), de Regt and Dieks (2005) distinguish three levels of analysis of the scientific community:
- *The **macro-level** of science as a whole; the **meso-level** of the scientific communities; and the **micro-level** of individual scientists... The three-level distinction reconciles the existence of **universal aims of science** with the existence of **variation** in the precise specification and/or application of these general aims*
 - Understanding is a macro-level (i.e. universal) aim of science
- Standards of intelligibility not universally determined for all of science: variation across scientific communities, i.e. at the meso-level of a discipline
 - Objective but **relative to level of progress** of a given discipline i.e. contextual

DE REGT'S CONTEXTUAL THEORY OF UNDERSTANDING

- “Grammar” of understanding:
- *Scientist **S** (in context **C**) understands phenomenon **P** on the basis of theory **T***
 - Understanding is pragmatic (relation subject/context)
 - Contextuality of understanding: variation is possible

Model-based explanation: $T \sim M \sim P$

- **M** represents **P** such that **T** can be applied to it
- No algorithms and strict rules for building models
- Instead: approximation and idealization, pragmatic decisions

S needs *skills* for constructing **M** to explain **P**

Understanding = skill to use theory for building models to explain phenomena

EXPLAINING PHENOMENA REQUIRES INTELLIGIBLE THEORIES

If **S** wants to explain a phenomenon on the basis of **T**,
she needs appropriate skills to use **T**

T should be *intelligible* to **S**

Intelligibility = the value that scientists attribute to the
qualities of a theory **T** that facilitate the use of the
theory.

- Not an intrinsic property of theories, but a context-dependent value related to scientists' skills
- Example: visualizability

UNDERSTANDING AND INTELLIGIBILITY

CUP: Criterion for Understanding Phenomena

- A phenomenon P is understood scientifically iff there is an explanation of P that is based on an intelligible theory T and conforms to the basic epistemic values of empirical adequacy and internal consistency.
- Scientists often understand phenomena by constructing models of them (including: simulations, etc.).

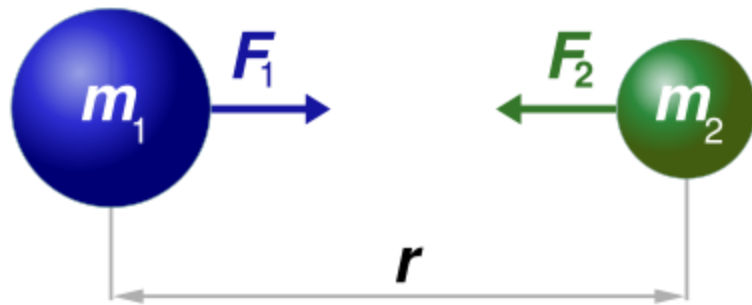
CIT : Criterion (test) for the Intelligibility of Theories

- A scientific theory T (in one or more of its representations) is intelligible for scientists (in context C) if they can **recognise qualitatively characteristic consequences of T without performing exact calculations**/fully explicit theoretical argumentation.
- The idea is that scientists have an “insight” into the workings of the theory and are accordingly able to use the theory for the construction of models of the phenomena, which satisfy the basic values of empirical adequacy and internal consistency.

HISTORICALLY DIFFERING STANDARDS OF INTELLIGIBILITY

- **Theories of gravitation:** from Newton (1687) to Einstein (1915)
 - Standards of intelligibility differ diachronically (i.e. in time, historically)
- **Quantum Theory** around 1926
 - Standards of intelligibility differ synchronically (i.e. regardless of time, for different scientists)

NEWTON'S THEORY OF GRAVITATION (1687)



Action at a distance



CHRISTIAAN HUYGENS ON GRAVITATION

'I look for an *understandable* cause of gravitation, because it seems to me that to say that bodies fall down because of some gravitational attraction—of earth or of those bodies—is to **say nothing**.'

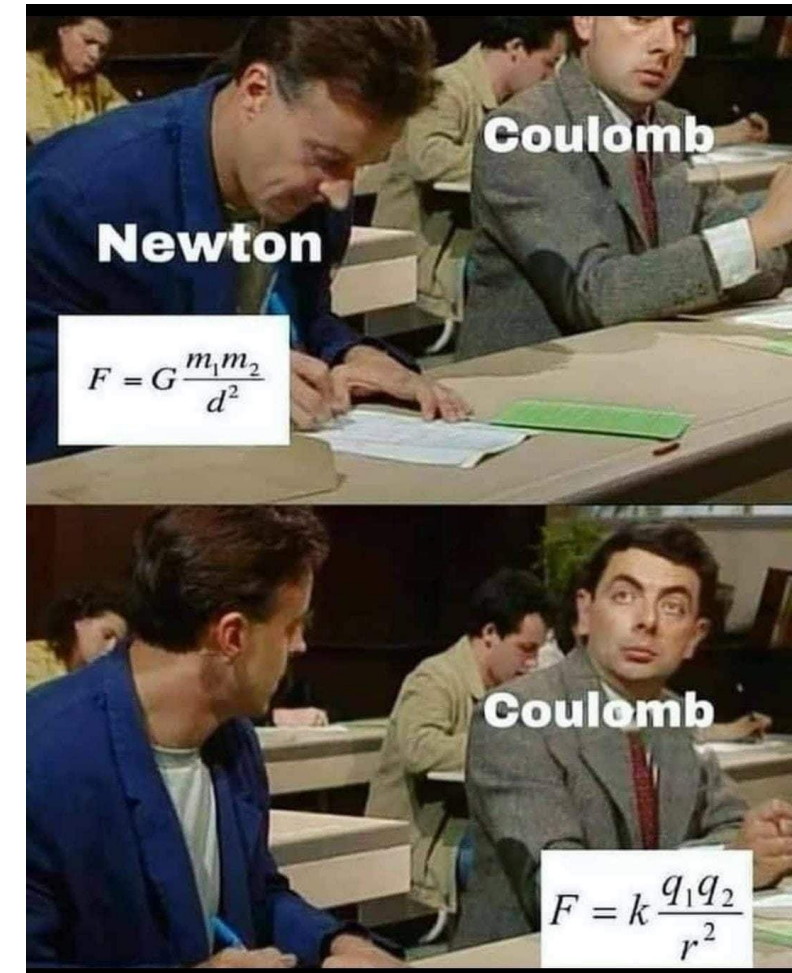


UNDERSTANDING AROUND 1800

- *Action at a distance* becomes the ideal of understanding
- Example: Coulomb's law

After 1850:

- Action by contact acceptable (Maxwell)
- Einstein's theories of relativity (1905 and 1915)



EXAMPLE 2: QUANTUM MECHANICS

Around 1926: competing theories of the atom

- Matrix mechanics (Heisenberg, Pauli): abstract
- Wave mechanics (Schrödinger): visualizable

Schrödinger vs. Pauli and Heisenberg debate

about the question of when a theory is visualizable (*anschaulich*).

SCHRÖDINGER AND *ANSCHAULICHKEIT*

Schrödinger on Heisenberg:

'I naturally knew about his theory, but was discouraged, if not repelled, by what appeared to me as very difficult methods of transcendental algebra, and by the lack of *Anschaulichkeit*.'

Anschaulichkeit: visualisability, intelligibility.

Physicists indeed found Schrödinger's theory easier to deal with, and so it was more widely used.

SCHRÖDINGER ON SCIENTIFIC UNDERSTANDING

‘We cannot really alter our manner of thinking in space and time, and what we cannot comprehend within it we cannot understand at all.’

Erwin Schrödinger (1926)



PAULI ON PICTURES AND ANSCHAULICHKEIT

‘.... our good friend Kramers and his colorful picture books – ‘and the children, they love to listen.’

Even though the demand of these children for visualizability (*Anschaulichkeit*) is partly legitimate and healthy, this should never count as an argument for the retention of fixed conceptual systems in physics. Once the new conceptual systems are settled, then also these will be *anschaulich*.’

Wolfgang Pauli (1924)



1920-1925: LOSS OF VISUALIZABILITY

Wave-particle duality

of light (Einstein, 1905)
and matter (De Broglie, 1923)

- no unambiguous visualization,
- no particle trajectories

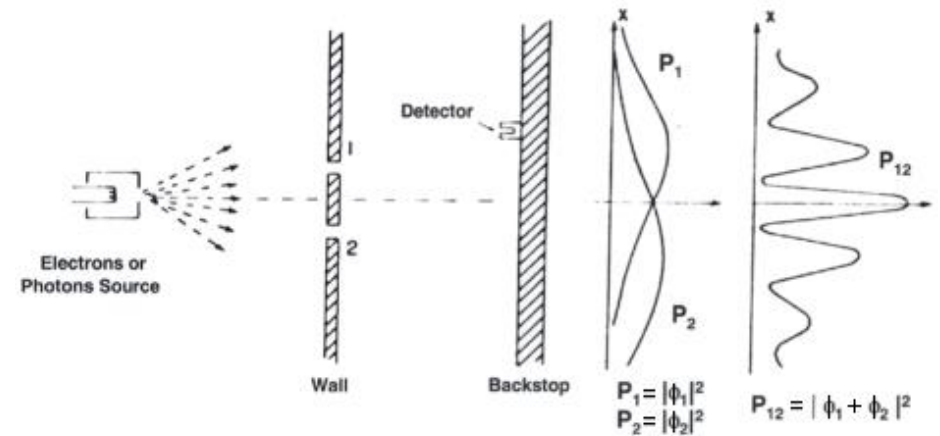


Figure 2. The double-slit experiment.

Reality of electron orbits disputed

- Pauli's fourth quantum number
- Atoms now completely non-visualizable.

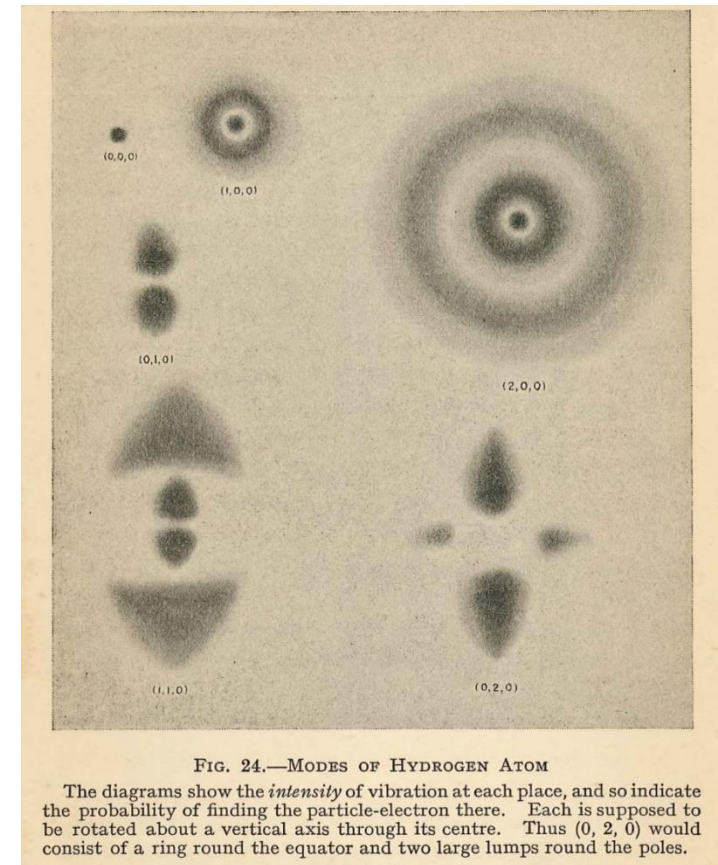
Search for a radically new '**quantum mechanics**' (Born)

VISUALIZING THE HYDROGEN ATOM

Representations of simple solutions of the hydrogen wave function

- Schrödinger's realistic interpretation (as charge densities) fails
- Instead: to be interpreted as probability densities.

From C.G. Darwin, *The New Conception of Matter* (1931)



OUTCOME OF THE DEBATE

- Schrödinger's visualization was problematic
- Heisenberg went on to use visualizable concepts
- Result: a new *quantum mechanics* that combined both theories (of course also other ingredients: Dirac's mathematical unification, Born's interpretation).

Visualizability in this historical context valued as a property that increases the **intelligibility** of theories.

- Context-dependence: visualizability not necessary for understanding.

SUMMARY

- Hempel's **deductive-nomological model of explanation**
 - Closely connected with **prediction from laws**
 - Objections: asymmetry, irrelevance, expectation/correlation
- Alternative models:
 - Causal-mechanistic explanation
 - Unification
- **Understanding** as knowledge/explanation and as more than explanation
- **Pragmatic theories: explanation + use/abilities**
 - Scientific understanding of phenomena requires intelligible theories.
- **Intelligibility** is the value that scientists **S**, in a context **C**, ascribe to the properties of a theory **T**, and which facilitates the use of this theory.
 - **Contextual:** as illustrated by visualizability/Anschaulichkeit of quantum mechanics

REFERENCES

- J. Woodward, L. Ross, *Scientific Explanation*, SEP, <https://plato.stanford.edu/entries/scientific-explanation>
- Reutlinger et al. (2018). 'Understanding (with) Toy Models'. *British Journal for the Philosophy of Science*, <https://www.journals.uchicago.edu/doi/full/10.1093/bjps/axx005>
- De Regt, H. (2017). *Understanding Scientific Understanding*. Oxford University Press